

EXECUTIVE BRIEF

Medicines Before Stock-Out

What leaders should authorize, require, and refuse

Production Unit	PU-B07-C25
Decision status	Controlled learning asset
Audience	Executives, programme owners, statisticians, technical leads
Data	Synthetic examples only

Executive takeaway: Anticipate medicine and vaccine stock-outs using demand, lead time, cold-chain, inventory quality, uncertainty, and governed replenishment rules.

Why it matters

Inventory totals can overstate usable supply when products are expired, inaccessible, misclassified, or cold-chain compromised.

Illustrative application

A vaccine alert combines usable stock, consumption, wastage, lead time, storage capacity, and expiry dates.

Minimum approval conditions

- Named accountable owner and lawful purpose
- Defined evidence, provenance, reference period, population, and geography
- Decision-relevant validation and visible uncertainty
- Documented abstention, referral, override, and incident processes
- Accessibility, equity, security, retention, and retirement review

Non-negotiable boundary

The analytical system may inform review. It may not independently authorize benefits, sanctions, enforcement, clinical treatment, school exclusion, displacement, or another consequential action.